

Test 17: Connes self-duality ??1/?

Verdict: PASS | Generated: 2026-09-15 00:55:45 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT: isospectrality under ? ? 1/? to machine precision.

METHOD: EXACT: isospectrality under ? ? 1/? to machine precision.

Test 17 HARDCORE L=24: zero_modes(?=1/2)=4, spectral defects 1.42e-15/1.66e-15, duality defect 3.62e-01.

Test 17 HARDCORE L=48: zero_modes(?=1/2)=4, spectral defects 2.90e-15/2.41e-15, duality defect 3.65e-01.

Test 17 HARDCORE L=96: zero_modes(?=1/2)=4, spectral defects 3.96e-15/3.24e-15, duality defect 3.66e-01.

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- ‘plots/plot_01.png’ ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- ‘plots/animation.gif’ ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 17 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[00:32:11.030] [+2722.058s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:32:11.825] [+2722.853s] heevr values-only solve: N=576 (?=0.5000, torus, W=0) ? zero_modes + E?-E defect ????? ????????

FULL COMPUTATION LOG

[00:32:11.976] [+2723.006s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:32:12.540] [+2723.568s] heevr values-only solve: N=576 (?=0.3333, torus, W=0) ? spectral_symmetry_defect

[00:32:12.697] [+2723.725s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:32:13.231] [+2724.259s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:32:13.780] [+2724.808s] heevr values-only solves x2: N=576 (?=0.5000 open, W=4) + N=576 (?=2.0000 open, W=4) ? connes_self_duality_defect

[00:32:14.746] [+2725.775s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:32:15.470] [+2726.498s] heevr values-only solve: N=2304 (?=0.5000, torus, W=0) ? zero_modes + E?-E defect ????? ????????

[00:32:20.921] [+2731.950s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:32:21.733] [+2732.761s] heevr values-only solve: N=2304 (?=0.3333, torus, W=0) ? spectral_symmetry_defect

[00:32:27.105] [+2738.133s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:32:27.927] [+2738.955s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:32:28.702] [+2739.730s] heevr values-only solves x2: N=2304 (?=0.5000 open, W=4) + N=2304 (?=2.0000 open, W=4) ? connes_self_duality_defect

[00:32:40.319] [+2751.347s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:32:47.777] [+2758.805s] heevr values-only solve: N=9216 (?=0.5000, torus, W=0) ? zero_modes + E?-E defect ????? ????????

[00:38:24.371] [+3095.399s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[00:38:35.596] [+3106.625s] heevr values-only solve: N=9216 (?=0.3333, torus, W=0) ? spectral_symmetry_defect

[00:44:36.057] [+3467.085s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:44:49.823] [+3480.852s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[00:45:02.209] [+3493.237s] heevr values-only solves $\times 2$: N=9216 (?=0.5000 open, W=4) + N=9216 (?=2.0000 open, W=4) ? connes_self_duality_defect

CONSOLE CAPTURE

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TEST 17 (pass 2, HARDCORE): Connes self-duality ? cross-size audit

Zero-mode invariant + spectral symmetries on the size ladder 24 ? ab_nx

??

L=24: zero_modes(?=1/2)=4 | spec.defect(1/2)=1.42e-15, (1/3)=1.66e-15 | duality=3.62e-01

L=48: zero_modes(?=1/2)=4 | spec.defect(1/2)=2.90e-15, (1/3)=2.41e-15 | duality=3.65e-01

L=96: zero_modes(?=1/2)=4 | spec.defect(1/2)=3.96e-15, (1/3)=3.24e-15 | duality=3.66e-01

At ?=1/3 the spectrum stays gapped at E=0 on these tori (base observation).

??

zero_modes(1/2)=4 ?L PASS ? topological invariant holds on the whole ladder

E??E defect < 1e-10 ?L PASS ? ?=1/2 and ?=1/3, generic bipartite symmetry

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS